

Variable Listing -- OCP Maintenance AI MVP

Generated from source code -- `tools/models/schemas.py` , `api/database/models.py` , `tools/engines/` , `api/routers/` , and `templates/generate_templates.py` .

Covers all enumerations, database columns, API endpoint fields, KPI definitions, the 72 valid Failure Mode combinations, and all 14 Excel template field schemas.

Section A: Enumeration Types

All `str`, `Enum` classes defined in `tools/models/schemas.py` , organized by domain.

A.1 Equipment & Hierarchy Domain

A.1.1 `EquipmentCriticality`

OCP criticality grades for equipment ranking.

Value	Description
<code>AA</code>	Double-A -- highest criticality
<code>A+</code>	A-plus -- very high criticality
<code>A</code>	A-grade criticality
<code>B</code>	B-grade criticality
<code>C</code>	C-grade criticality
<code>D</code>	D-grade -- lowest criticality

A.1.2 `EquipmentStatus`

Lifecycle state of a physical asset.

Value	Description
<code>ACTIVE</code>	Equipment is in operation
<code>INACTIVE</code>	Equipment is idle / mothballed
<code>DECOMMISSIONED</code>	Equipment permanently removed

A.1.3 `NodeType`

Level type in the 6-level plant hierarchy.

Value	Level	Description
PLANT	1	Top-level facility
AREA	2	Functional area within plant
SYSTEM	3	Process system
EQUIPMENT	4	Physical equipment item
SUB_ASSEMBLY	5	Sub-assembly of equipment
MAINTAINABLE_ITEM	6	Lowest replaceable component

A.1.4 EquipmentCategory

Classification of equipment by type (Module 4 library).

Value	Description
PUMP	Centrifugal, positive displacement, etc.
CONVEYOR	Belt, screw, chain conveyors
CRUSHER	Jaw, cone, gyratory crushers
MILL	SAG, ball, rod mills
FLOTATION_CELL	Flotation separation
THICKENER	Thickening/settling
DRYER	Drying equipment
KILN	Rotary kilns
MOTOR	Electric motors
TRANSFORMER	Electrical transformers
COMPRESSOR	Gas compressors
VALVE	Control / isolation valves
TANK	Storage / process tanks
FILTER	Filtration equipment
AGITATOR	Agitation equipment
OTHER	Uncategorised

A.1.5 ComponentCategory

Component classification by discipline.

Value	Description
MECHANICAL	Mechanical components
ELECTRICAL	Electrical components
INSTRUMENTATION	Sensors, PLCs, transmitters
STRUCTURAL	Frames, housing, supports
HYDRAULIC	Hydraulic systems
PNEUMATIC	Pneumatic systems

A.1.6 LibrarySource

Origin of a library item.

Value	Description
R8_LIBRARY	From R8 standard library
OEM	From OEM documentation
CUSTOM	User-defined
AI_GENERATED	Generated by AI engine

A.2 Field Capture Domain

A.2.1 CaptureType

Type of field data capture.

Value	Description
VOICE	Voice recording transcribed
TEXT	Typed text input
IMAGE	Photo capture only
VOICE+IMAGE	Combined voice + photo

A.2.2 Language

Supported languages for field capture.

Value	Description
fr	French
en	English
ar	Arabic

A.2.3 ResolutionMethod

How equipment was identified from field input.

Value	Description
EXACT_MATCH	Exact TAG/ID match
FUZZY_MATCH	Fuzzy text matching
IMAGE_OCR	OCR from photo
MANUAL	Manually selected

A.2.4 VisualSeverity

Severity from image analysis.

Value	Description
LOW	Minor visual anomaly
MEDIUM	Moderate degradation
HIGH	Significant damage
CRITICAL	Immediate attention required

A.3 Work Orders Domain

A.3.1 WorkRequestStatus

Lifecycle of a structured work request.

Value	Description
DRAFT	Initial creation
PENDING_VALIDATION	Awaiting planner review
VALIDATED	Approved by planner
REJECTED	Rejected by planner
SUBMITTED_TO_SAP	Sent to SAP PM

A.3.2 WorkOrderType

SAP PM order types.

Value	Description
PM01_INSPECTION	Inspection order
PM02_PREVENTIVE	Preventive maintenance
PM03_CORRECTIVE	Corrective maintenance

A.3.3 Priority

Work order priority levels.

Value	Description
1_EMERGENCY	Immediate response
2_URGENT	Response within shift
3_NORMAL	Planned within week
4_PLANNED	Scheduled per plan

A.3.4 AvailabilityStatus

Spare part stock status.

Value	Description
IN_STOCK	Available in warehouse
LOW_STOCK	Below reorder point
OUT_OF_STOCK	Zero on hand
UNKNOWN	Status not determined

A.3.5 BacklogWOType

Simplified SAP order type for backlog.

Value	Description
PM01	Inspection
PM02	Preventive
PM03	Corrective

A.3.6 BacklogStatus

Status of a backlog item.

Value	Description
AWAITING_MATERIALS	Blocked on parts
AWAITING_SHUTDOWN	Needs shutdown window
AWAITING_RESOURCES	No available workforce
AWAITING_APPROVAL	Pending authorization
SCHEDULED	Placed on schedule
IN_PROGRESS	Execution underway

A.3.7 WOHistoryStatus

Work order lifecycle states (historical).

Value	Description
CREATED	WO created in system
RELEASED	WO released for scheduling
IN_PROGRESS	Execution started
COMPLETED	Work finished
CLOSED	Administratively closed
CANCELLED	WO cancelled

A.3.8 MaterialsReadyStatus

Material readiness for a work package.

Value	Description
READY	All parts available
PARTIAL	Some parts missing
NOT_READY	Parts not available

A.3.9 AlertType

Backlog alert categories.

Value	Description
OVERDUE	Item past due date
MATERIAL_DELAY	Material delivery delayed
RESOURCE_CONFLICT	Workforce double-booked
PRIORITY_ESCALATION	Priority has been raised

A.3.10 SAPWorkOrderStatus

SAP PM work order system statuses.

Value	Description
PLN	Planned
FMA	Firm
LPE	Released for execution
LIB	Released
IMPR	In process
NOTP	Not planned
NOTI	Notification
CTEC	Technically completed

A.3.11 SAPNotificationStatus

SAP PM notification statuses.

Value	Description
MEAB	Open
METR	In process
ORAS	Order assigned
MECE	Closed

A.4 Planner & Scheduling Domain

A.4.1 ShiftType

Shift definitions.

Value	Description
MORNING	Day shift
AFTERNOON	Swing shift
NIGHT	Night shift

A.4.2 ShutdownType

Planned shutdown categories.

Value	Description
MINOR_8H	Minor -- up to 8 hours
MAJOR_20H_PLUS	Major -- 20+ hours

A.4.3 RiskLevel

General risk classification.

Value	Description
LOW	Acceptable risk
MEDIUM	Monitor and plan
HIGH	Action required
CRITICAL	Immediate intervention

A.4.4 PlannerAction

Actions a planner can take on a recommendation.

Value	Description
APPROVE	Accept as-is
MODIFY	Change details and approve
ESCALATE	Raise to management
DEFER	Postpone execution

A.4.5 PlanStrategy

Maintenance plan strategy type.

Value	Description
TIME_BASED	Calendar/meter interval
CONDITION_BASED	Triggered by measurements
PREDICTIVE	Data-driven prediction

A.4.6 FrequencyUnit

Units for maintenance task frequency.

Value	Description
HOURS	Calendar hours
DAYS	Calendar days
WEEKS	Calendar weeks
MONTHS	Calendar months
YEARS	Calendar years
HOURS_RUN	Running hours
OPERATING_HOURS	Operating hours
TONNES	Throughput in tonnes
CYCLES	Machine cycles

A.4.7 PlanStatus

Maintenance plan lifecycle.

Value	Description
ACTIVE	Currently in effect
SUSPENDED	Temporarily paused
DELETED	Permanently removed

A.4.8 WeeklyProgramStatus

Weekly scheduling program states.

Value	Description
DRAFT	Being assembled
FINAL	Locked for execution
ACTIVE	Currently executing
COMPLETED	Execution finished

A.4.9 SchedulingDayRole

Purpose of each day in the weekly scheduling cycle.

Value	Description
MEETING	Scheduling coordination meeting
ADJUSTMENT	Mid-week adjustments
FINAL_PROGRAM	Final program freeze
EXECUTION	Execution day

A.4.10 ShutdownScope

Scope classification for shutdown windows.

Value	Description
MINOR_8H	Minor -- up to 8 hours
MAJOR_20H_PLUS	Major -- 20+ hours

A.4.11 SupportTaskType

Support activities required alongside work packages.

Value	Description
LOTO	Lock-out/Tag-out
SCAFFOLDING	Scaffolding erection/removal
CRANE	Crane mobilization
MANLIFT	Manlift requirement
GUARD_REMOVAL	Guard removal/reinstallation
CLEANING	Area cleaning
COMMISSIONING	Post-work commissioning

A.4.12 WorkPackageElementType

Required elements of a complete work package.

Value	Description
WORK_PERMIT	Work permit document
LOTO_CERT	LOTO certificate
MATERIAL_WITHDRAWAL	Material issue slip
CHECKLIST	Pre-job checklist
ATS_ART	Risk analysis (ATS/ART)
EXECUTION_PROCEDURE	Step-by-step procedure
WORK_ORDER	SAP work order reference

A.5 Criticality & FMEA Domain

A.5.1 CriticalityMethod

Assessment methodology selector.

Value	Description
FULL_MATRIX	Full 11-category risk matrix
SIMPLIFIED	Simplified assessment

A.5.2 CriticalityCategory

Risk consequence categories (R8 method).

Value	Description
SAFETY	Personnel safety impact
HEALTH	Occupational health impact
ENVIRONMENT	Environmental impact
PRODUCTION	Production loss
OPERATING_COST	Operating cost increase
CAPITAL_COST	Capital expenditure
SCHEDULE	Schedule impact
REVENUE	Revenue loss
COMMUNICATIONS	Communication systems
COMPLIANCE	Regulatory compliance
REPUTATION	Company reputation

A.5.3 RiskClass

Resulting risk classification from assessment.

Value	Description
I_LOW	Class I -- Low risk
II_MEDIUM	Class II -- Medium risk
III_HIGH	Class III -- High risk
IV_CRITICAL	Class IV -- Critical risk

A.5.4 ApprovalStatus

Approval workflow for strategy items.

Value	Description
DRAFT	Initial state
REVIEWED	Peer-reviewed
APPROVED	Management-approved

A.5.5 FunctionType

Equipment function classification (RCM).

Value	Description
PRIMARY	Main intended function
SECONDARY	Supporting function
PROTECTIVE	Safety/protection function

A.5.6 FailureType

Functional failure degree.

Value	Description
TOTAL	Complete loss of function
PARTIAL	Degraded performance

A.5.7 FMStatus

Failure mode recommendation status.

Value	Description
RECOMMENDED	Active and recommended
REDUNDANT	Duplicate / not needed

A.5.8 CriticalityMode

GFSN vs R8 assessment mode selector.

Value	Description
R8	R8 standard method
GFSN	GFSN OCP method

A.5.9 GFSNCriticalityBand

GFSN criticality classification bands.

Value	Description
ALTO	High criticality
MODERADO	Moderate criticality
BAJO	Low criticality

A.5.10 GFSNConsequenceCategory

GFSN consequence assessment dimensions.

Value	Description
BUSINESS_IMPACT	Business impact
OPERATIONAL_COST	Operational cost impact
INTERRUPTION	Service interruption
SAFETY	Safety impact
ENVIRONMENT	Environmental impact
RSC	Regulatory / Social / Community

A.5.11 PriorityMode

Priority calculation method selector.

Value	Description
ADDITIVE	Simple additive scoring
GFSN_MATRIX	GFSN matrix method

A.5.12 GFSNPriority

GFSN priority levels.

Value	Description
ALTO	High priority
MODERADO	Moderate priority
BAJO	Low priority

A.6 FMEA / RCM Domain

A.6.1 Mechanism

18 valid failure mechanisms from SRC-09.

Value	Description
ARCS	Electrical arcing
BLOCKS	Blockage / obstruction
BREAKS_FRACTURE_SEPARATES	Fracture or mechanical break
CORRODES	Corrosion degradation
CRACKS	Crack propagation
DEGRADES	General degradation
DISTORTS	Physical distortion
DRIFTS	Parameter drift
EXPIRES	End-of-life expiry
IMMOBILISED	Seizure / immobilization
LOOSES_PRELOAD	Loss of preload / fastener loosening
OPEN_CIRCUIT	Electrical open circuit
OVERHEATS_MELTS	Overheating or melting
SEVERS	Severing / cutting
SHORT_CIRCUITS	Electrical short circuit
THERMALLY_OVERLOADS	Thermal overload
WASHES_OFF	Wash-off / erosion
WEARS	Mechanical wear

A.6.2 Cause

41 valid failure causes from SRC-09.

Value	Description
ABRASION	Abrasive wear
AGE	Age-related deterioration
BREAKDOWN_IN_INSULATION	Insulation breakdown
BREAKDOWN_OF_LUBRICATION	Lubrication failure
BIO_ORGANISMS	Biological organisms
CHEMICAL_ATTACK	Chemical attack
CHEMICAL_REACTION	Chemical reaction
CONTAMINATION	Contamination ingress
CORROSIVE_ENVIRONMENT	Corrosive environment exposure
CREEP	Material creep
CREVICE	Crevice corrosion
CYCLIC_LOADING	Cyclic fatigue loading
DISSIMILAR_METALS_CONTACT	Galvanic corrosion
ELECTRICAL_ARCING	Electrical arcing damage
ELECTRICAL_OVERLOAD	Electrical overload
ENTRAINED_AIR	Entrained air / cavitation
EXCESSIVE_FLUID_VELOCITY	Excessive fluid velocity
EXCESSIVE_PARTICLE_SIZE	Oversized particles
EXCESSIVE_TEMPERATURE	Excessive temperature
EXPOSURE_TO_ATMOSPHERE	Atmospheric exposure
EXPOSURE_TO_HIGH_TEMP_CORROSIVE	High-temperature corrosive exposure
EXPOSURE_TO_HIGH_TEMP	High-temperature exposure
EXPOSURE_TO_LIQUID_METAL	Liquid metal contact
HIGH_TEMP_CORROSIVE	High-temperature corrosion
IMPACT_SHOCK_LOADING	Impact / shock load
INSUFFICIENT_FLUID_VELOCITY	Insufficient fluid velocity
LACK_OF_LUBRICATION	Lubrication starvation
LOW_PRESSURE	Low fluid pressure
LUBRICANT_CONTAMINATION	Contaminated lubricant
MECHANICAL_OVERLOAD	Mechanical overload
METAL_TO_METAL_CONTACT	Dry metal contact
OFF_CENTER_LOADING	Misaligned / off-center load
OVERCURRENT	Overcurrent condition
POOR_ELECTRICAL_CONNECTIONS	Poor electrical connections

Value	Description
POOR_ELECTRICAL_INSULATION	Poor electrical insulation
RADIATION	Radiation damage
RELATIVE_MOVEMENT	Relative movement / fretting
RUBBING	Rubbing contact
STRAY_CURRENT	Stray electrical current
THERMAL_OVERLOAD	Thermal overload
THERMAL_STRESSES	Thermal stress cycling
UNEVEN_LOADING	Uneven loading
USE	Normal use / wear
VIBRATION	Vibration-induced loosening

A.6.3 FailurePattern

RCM failure-rate curves (Nowlan-Heap).

Value	Description
A_BATHTUB	Pattern A -- bathtub curve (infant + wearout)
B_AGE	Pattern B -- age-related wearout
C_FATIGUE	Pattern C -- gradual fatigue increase
D_STRESS	Pattern D -- initial stress then flat
E_RANDOM	Pattern E -- constant random failure
F_EARLY_LIFE	Pattern F -- infant mortality then flat

A.6.4 FailureConsequence

RCM consequence classification.

Value	Description
HIDDEN_SAFETY	Hidden failure with safety impact
HIDDEN_NONSAFETY	Hidden failure without safety impact
EVIDENT_SAFETY	Evident failure with safety impact
EVIDENT_ENVIRONMENTAL	Evident failure with environmental impact
EVIDENT_OPERATIONAL	Evident failure with production impact
EVIDENT_NONOPERATIONAL	Evident failure -- no direct impact

A.6.5 StrategyType

RCM maintenance strategy selection.

Value	Description
CONDITION_BASED	Condition monitoring / on-condition task
FIXED_TIME	Fixed interval replacement / overhaul
RUN_TO_FAILURE	No planned maintenance
FAULT_FINDING	Periodic hidden-function check
REDESIGN	Design change required
OEM	Follow OEM recommendation

A.6.6 FMECAStage

Progressive FMECA worksheet stages.

Value	Description
STAGE_1_FUNCTIONS	Define functions
STAGE_2_FAILURES	Identify functional failures
STAGE_3_EFFECTS	Analyse failure effects
STAGE_4_DECISIONS	RCM decision logic

A.6.7 FMECAWorksheetStatus

FMECA worksheet lifecycle.

Value	Description
DRAFT	Being developed
IN_PROGRESS	Analysis underway
COMPLETED	Analysis complete
APPROVED	Reviewed and approved

A.6.8 RPNCategory

Risk Priority Number classification (FMECA).

Value	Description
LOW	RPN in low range
MEDIUM	RPN in medium range
HIGH	RPN in high range
CRITICAL	RPN in critical range

A.6.9 JustificationCategory

Justification for strategy changes vs. existing plans.

Value	Description
MODIFIED	Existing task modified
ELIMINATED	Existing task removed
FREQUENCY_CHANGE	Frequency adjusted
TACTIC_CHANGE	Strategy/tactic changed
MAINTAINED	Kept as-is
NEW_TASK	Brand new task added

A.7 Task & Work Package Domain

A.7.1 TaskType

Maintenance task classifications.

Value	Description
INSPECT	Visual / sensory inspection
CHECK	Functional check / measurement
TEST	Diagnostic test
LUBRICATE	Lubrication task
CLEAN	Cleaning task
REPLACE	Component replacement
REPAIR	Repair activity
CALIBRATE	Instrument calibration

A.7.2 TaskConstraint

Task execution constraint.

Value	Description
ONLINE	Can run while equipment operates
OFFLINE	Requires equipment shutdown
TEST_MODE	Requires test / bypass mode

A.7.3 LabourSpecialty

Workforce trade specialty.

Value	Description
FITTER	Mechanical fitter
ELECTRICIAN	Electrician
INSTRUMENTIST	Instrument technician
OPERATOR	Plant operator
COMMON_SPECIALIST	Condition monitoring specialist
LUBRICATOR	Lubrication technician

A.7.4 UnitOfMeasure

Material quantity units.

Value	Description
EA	Each (unit)
L	Litre
KG	Kilogram
M	Metre

A.7.5 BudgetType

Budget classification for tasks.

Value	Description
REPAIR	Repair budget
REPLACE	Replacement budget

A.7.6 WPType

Work package grouping strategy.

Value	Description
STANDALONE	Single independent package
SUPPRESSIVE	Grouped to suppress shutdown count
SEQUENTIAL	Must execute in sequence

A.7.7 WPConstraint

Work package shutdown requirement.

Value	Description
ONLINE	No shutdown required
OFFLINE	Shutdown required

A.7.8 WPApprovalStatus

Work package approval lifecycle.

Value	Description
DRAFT	Being assembled
REVIEWED	Peer-reviewed
APPROVED	Management-approved
UPLOADED_TO_SAP	Transferred to SAP PM

A.8 SAP Integration Domain

A.8.1 SAPUploadStatus

SAP upload package lifecycle.

Value	Description
GENERATED	Package generated by system
REVIEWED	Reviewed by planner
APPROVED	Approved for upload
UPLOADED	Successfully uploaded to SAP

A.9 Spare Parts & Reliability Domain

A.9.1 MaterialCriticality

Spare part criticality classification.

Value	Description
CRITICAL	Failure causes immediate stoppage
IMPORTANT	Failure degrades performance
STANDARD	Routine consumable

A.9.2 SparePartCriticality

VED analysis classification.

Value	Description
VITAL	Production stops without this part
ESSENTIAL	Significant impact if unavailable
DESIRABLE	Nice-to-have in stock

A.9.3 ConsumptionClass

FSN inventory velocity classification.

Value	Description
FAST_MOVING	High consumption rate
SLOW_MOVING	Low consumption rate
NON_MOVING	No movement in review period

A.9.4 CostClass

ABC cost-based inventory classification.

Value	Description
A_HIGH	High-value items (~20% SKUs, ~80% value)
B_MEDIUM	Medium-value items
C_LOW	Low-value items (~80% SKUs, ~20% value)

A.9.5 ShutdownStatus

Shutdown event lifecycle.

Value	Description
PLANNED	Scheduled for future
IN_PROGRESS	Currently executing
COMPLETED	Successfully finished
CANCELLED	Cancelled

A.9.6 MoCStatus

Management of Change request lifecycle.

Value	Description
DRAFT	Being prepared
SUBMITTED	Submitted for review
REVIEWING	Under review
APPROVED	Approved
IMPLEMENTING	Being implemented
CLOSED	Implementation verified
REJECTED	Rejected

A.9.7 MoCCategory

Management of Change classification.

Value	Description
EQUIPMENT_MODIFICATION	Physical equipment change
PROCESS_CHANGE	Process parameter change
STRATEGY_CHANGE	Maintenance strategy change
MATERIAL_SUBSTITUTION	Alternative material/part
PROCEDURE_UPDATE	Procedure revision

A.9.8 JackKnifeZone

Jack-Knife diagram zone classification (REF-13 7.5.4).

Value	Description
ACUTE	High frequency, high downtime per event
CHRONIC	High frequency, low downtime per event
COMPLEX	Low frequency, high downtime per event
CONTROLLED	Low frequency, low downtime per event

A.9.9 DamageMechanism

RBI damage mechanism classification.

Value	Description
CORROSION	General or localised corrosion
FATIGUE	Cyclic fatigue
CREEP	High-temperature creep
EROSION	Erosive wear
STRESS_CORROSION	Stress corrosion cracking
HYDROGEN_DAMAGE	Hydrogen embrittlement
OTHER	Other mechanisms

A.9.10 InspectionTechnique

RBI recommended inspection methods.

Value	Description
VISUAL	Visual inspection
ULTRASONIC_THICKNESS	UT thickness measurement
MAGNETIC_PARTICLE	MPI
DYE_PENETRANT	DPI / LPI
RADIOGRAPHY	Radiographic testing
EDDY_CURRENT	Eddy current testing
ACOUSTIC_EMISSION	Acoustic emission monitoring

A.10 KPI & Governance Domain

A.10.1 KPIStatus

KPI target attainment status.

Value	Description
ON_TARGET	Meeting or exceeding target
BELOW_TARGET	Below target
ABOVE_TARGET	Above target (may be negative for lower-is-better)

A.10.2 CAPAStatus

CAPA lifecycle (ISO 55002 10.2).

Value	Description
OPEN	Newly created
IN_PROGRESS	Actions underway
CLOSED	Actions completed
VERIFIED	Effectiveness verified

A.10.3 CAPAType

Corrective vs Preventive action.

Value	Description
CORRECTIVE	Corrective action
PREVENTIVE	Preventive action

A.10.4 PDCAPhase

Deming PDCA cycle phases.

Value	Description
PLAN	Planning phase
DO	Execution phase
CHECK	Verification phase
ACT	Standardization phase

A.10.5 HealthDimension

5 dimensions of the Asset Health Index (REF-12 Rec 4).

Value	Description
CRITICALITY	Equipment criticality assessment
BACKLOG_PRESSURE	Ratio of pending work to capacity
STRATEGY_COVERAGE	% failure modes with approved strategy
CONDITION_STATUS	Active condition monitoring alerts
EXECUTION_COMPLIANCE	On-time execution of planned work

A.10.6 VarianceLevel

Multi-plant variance alert severity (REF-12 Rec 7).

Value	Description
NORMAL	Within 2 sigma of portfolio mean
WARNING	Between 2-3 sigma deviation
CRITICAL	Beyond 3 sigma deviation

A.10.7 ExpertDomain

Expert specialization domains (Neuro-Arq TMS).

Value	Description
RELIABILITY	Reliability engineering
MECHANICAL	Mechanical maintenance
ELECTRICAL	Electrical maintenance
INSTRUMENTATION	Instrumentation & control
PROCESS	Process engineering
SAFETY	Safety engineering
PLANNING	Maintenance planning
SPARE_PARTS	Spare parts management

A.10.8 StakeholderRole

Stakeholder roles (ISO 55002 4.2).

Value	Description
MAINTENANCE_MANAGER	Head of maintenance
RELIABILITY_ENGINEER	Reliability specialist
PLANNER	Maintenance planner
TECHNICIAN	Field technician
OPERATOR	Plant operator
PLANT_MANAGER	Plant director
SAFETY_OFFICER	HSE officer
PROCUREMENT	Procurement / stores

A.10.9 RCALevel

RCA investigation depth level.

Value	Description
1	Level 1 -- Supervisor + operator
2	Level 2 -- Cross-functional team
3	Level 3 -- Full investigation with external support

A.10.10 RCAStatus

RCA analysis lifecycle.

Value	Description
OPEN	Newly opened
UNDER_INVESTIGATION	Investigation underway
COMPLETED	Analysis completed
REVIEWED	Final review complete

A.10.11 EvidenceType

RCA evidence classification.

Value	Description
INFERRED	Logically inferred
SENSORY	Directly observed
HYPOTHESIS	Hypothesized

A.10.12 RootCauseLevel

Root cause depth classification.

Value	Description
PHYSICAL	Physical root cause
HUMAN	Human error root cause
LATENT	Systemic / organizational

A.10.13 Evidence5PCategory

5P evidence collection categories.

Value	Description
PARTS	Physical parts evidence
POSITION	Location / position evidence
PEOPLE	Human witness evidence
PAPERS	Documentation evidence
PARADIGMS	Organizational / cultural patterns

A.10.14 SolutionQuadrant

Solution prioritization quadrant.

Value	Description
HIGH_BENEFIT_LOW_DIFFICULTY	Quick wins -- implement first
HIGH_BENEFIT_HIGH_DIFFICULTY	Strategic projects
LOW_BENEFIT_LOW_DIFFICULTY	Nice-to-have
LOW_BENEFIT_HIGH_DIFFICULTY	Avoid / deprioritize

A.11 Reporting & Dashboard Domain

A.11.1 ReportType

Generated report types.

Value	Description
WEEKLY_MAINTENANCE	Weekly maintenance summary
MONTHLY_KPI	Monthly KPI report
QUARTERLY_REVIEW	Quarterly management review

A.11.2 NotificationType

System alert notification types.

Value	Description
RBI_OVERDUE	Overdue RBI inspection
KPI_BREACH	KPI target breach
EQUIPMENT_RISK	Equipment risk escalation
BACKLOG_AGING	Aging backlog items
CAPA_OVERDUE	Overdue CAPA actions
MOC_OVERDUE	Overdue MoC requests

A.11.3 NotificationLevel

Alert severity.

Value	Description
INFO	Informational
WARNING	Warning
CRITICAL	Critical

A.11.4 ExportFormat

Data export format options.

Value	Description
EXCEL	Microsoft Excel (.xlsx)
CSV	Comma-separated values
PDF	PDF document

A.11.5 ImportSource

Import data source types.

Value	Description
EQUIPMENT_HIERARCHY	Equipment hierarchy import
FAILURE_HISTORY	Failure history import
MAINTENANCE_PLAN	Maintenance plan import

A.11.6 CorrelationType

Cross-module correlation analysis types.

Value	Description
CRITICALITY_FAILURES	Criticality vs failure frequency
COST_RELIABILITY	Cost vs reliability metrics
HEALTH_BACKLOG	Health score vs backlog pressure

A.11.7 TrafficLight

Dashboard traffic light indicators.

Value	Description
GREEN	On target
AMBER	Warning zone
RED	Off target

A.12 Phase 7 Domain

A.12.1 WorkPackageReadiness

Assembly readiness status.

Value	Description
NOT_STARTED	No elements assembled
PARTIAL	Some elements present
READY	All elements present
BLOCKED	Missing critical element

A.12.2 ElementReadinessStatus

Individual element readiness.

Value	Description
MISSING	Not yet created
DRAFT	Draft version available
READY	Final version available
EXPIRED	Expired, needs renewal

A.12.3 SupportTaskStatus

Support task progress.

Value	Description
PENDING	Not yet started
IN_PROGRESS	Underway
COMPLETED	Done
SKIPPED	Not required / skipped

A.12.4 ConflictResolutionType

Scheduling conflict resolution strategies.

Value	Description
RESCHEDULE	Move to different date/shift
SPLIT_PACKAGE	Break package into parts
ADD_SHIFT	Add extra shift
REASSIGN_SPECIALTY	Use alternate trade
EXTEND_WINDOW	Extend shutdown window

Section B: Database Fields

All SQLAlchemy table definitions from `api/database/models.py`.

B.1 plants

Field	Type	Constraints	Description
plant_id	String(50)	PK	SAP Plant code (e.g., <code>OCP-JFC1</code>)
name	String(200)	NOT NULL	Plant name (English)
name_fr	String(200)	default=""	Plant name (French)
name_ar	String(200)	default=""	Plant name (Arabic)
location	String(200)	default=""	Geographic location

B.2 hierarchy_nodes

Field	Type	Constraints	Description
node_id	String(50)	PK , default=uuid	Unique node identifier
node_type	String(30)	NOT NULL	NodeType enum value
name	String(200)	NOT NULL	Node name (English)
name_fr	String(200)	default=""	Node name (French)
code	String(100)	indexed	Technical code
parent_node_id	String(50)	FK hierarchy_nodes.node_id, nullable	Parent reference (self-referential)
level	Integer	NOT NULL	Hierarchy level (1-6)
plant_id	String(50)	FK plants.plant_id, nullable	Owning plant
tag	String(100)	nullable, indexed	Equipment TAG
criticality	String(10)	nullable	EquipmentCriticality value
status	String(30)	default="ACTIVE"	EquipmentStatus value
equipment_lib_ref	String(50)	nullable	Equipment library reference
component_lib_ref	String(50)	nullable	Component library reference
sap_func_loc	String(100)	nullable	SAP functional location
sap_equipment_nr	String(100)	nullable	SAP equipment number
order	Integer	default=1	Display order within parent
metadata_json	JSON	nullable	NodeMetadata as JSON

Indexes: `ix_hierarchy_nodes_parent`, `ix_hierarchy_nodes_level_type`

B.3 criticality_assessments

Field	Type	Constraints	Description
assessment_id	String(50)	PK , default=uuid	Assessment identifier
node_id	String(50)	FK hierarchy_nodes.node_id	Assessed node
assessed_at	DateTime	default=now	Assessment timestamp
assessed_by	String(100)	default="system"	Assessor name
method	String(30)	NOT NULL	FULL_MATRIX or SIMPLIFIED
criteria_scores	JSON	NOT NULL	list[CriteriaScore]
probability	Integer	NOT NULL	1-5 probability rating
overall_score	Float	default=0.0	Computed overall risk score
risk_class	String(20)	NOT NULL	RiskClass value
ai_suggested_class	String(20)	nullable	AI recommendation
ai_justification	Text	nullable	AI reasoning
status	String(20)	default="DRAFT"	ApprovalStatus value

B.4 functions

Field	Type	Constraints	Description
function_id	String(50)	PK , default=uuid	Function identifier
node_id	String(50)	FK hierarchy_nodes.node_id	Equipment node
function_type	String(20)	NOT NULL	PRIMARY, SECONDARY, PROTECTIVE
description	Text	NOT NULL	Function description (English)
description_fr	Text	default=""	Function description (French)
performance_standard	Text	nullable	Performance criteria
ai_generated	Boolean	default=False	AI-generated flag
status	String(20)	default="DRAFT"	ApprovalStatus value

B.5 functional_failures

Field	Type	Constraints	Description
failure_id	String(50)	PK , default=uuid	Failure identifier
function_id	String(50)	FK functions.function_id	Parent function
failure_type	String(20)	NOT NULL	TOTAL or PARTIAL
description	Text	NOT NULL	Failure description (English)
description_fr	Text	default=""	Failure description (French)

B.6 failure_modes

Field	Type	Constraints	Description
failure_mode_id	String(50)	PK , default=uuid	FM identifier
functional_failure_id	String(50)	FK functional_failures.failure_id	Parent failure
fm_status	String(20)	default="RECOMMENDED"	RECOMMENDED or REDUNDANT
what	String(200)	NOT NULL	FM description (starts uppercase)
mechanism	String(50)	NOT NULL	Mechanism enum value
cause	String(50)	NOT NULL	Cause enum value
failure_pattern	String(30)	nullable	Nowlan-Heap pattern (A-F)
failure_consequence	String(30)	NOT NULL	FailureConsequence value
is_hidden	Boolean	default=False	Hidden failure flag
failure_effect	JSON	nullable	FailureEffect as JSON
strategy_type	String(30)	NOT NULL	StrategyType value
ai_generated	Boolean	default=False	AI-generated flag
ai_confidence	Float	nullable	AI confidence (0-1)
existing_task_source	String(200)	nullable	Source of existing task
justification_category	String(30)	nullable	JustificationCategory value

Indexes: ix_fm_mechanism_cause

B.7 maintenance_tasks

Note: Tasks are standalone, reusable action definitions. They do NOT have frequency or acceptable limits -- those fields belong to the strategy level. Frequency and limits are defined when a task is linked to a failure mode via a maintenance strategy (see Template 14: 14_maintenance_strategy.xlsx). The frequency_value and frequency_unit fields in the database are retained for backward compatibility with work packages but are NOT populated during task-only data loading (Template 04).

Field	Type	Constraints	Description
<code>task_id</code>	String(50)	PK , default=uuid	Task identifier (reusable across strategies)
<code>failure_mode_id</code>	String(50)	FK failure_modes.failure_mode_id, nullable	Linked failure mode
<code>name</code>	String(72)	NOT NULL	Task name (max 72 chars for SAP)
<code>name_fr</code>	String(200)	default=""	Task name (French)
<code>task_type</code>	String(20)	NOT NULL	TaskType value
<code>is_secondary</code>	Boolean	default=False	Secondary task flag
<code>acceptable_limits</code>	Text	nullable	Acceptance criteria (strategy-level field)
<code>conditional_comments</code>	Text	nullable	Conditional notes (strategy-level field)
<code>consequences</code>	Text	default=""	Failure consequences
<code>justification</code>	Text	nullable	Task justification
<code>constraint</code>	String(20)	NOT NULL	ONLINE, OFFLINE, TEST_MODE
<code>access_time_hours</code>	Float	default=0.0	Access/setup time
<code>frequency_value</code>	Float	NOT NULL	Interval value (set at strategy/WP level)
<code>frequency_unit</code>	String(30)	NOT NULL	FrequencyUnit value (set at strategy/WP level)
<code>origin</code>	String(200)	nullable	Task origin reference
<code>budget_type</code>	String(20)	nullable	NOT_BUDGETED, REPAIR, or REPLACE
<code>budgeted_life</code>	Float	nullable	Budgeted life value
<code>labour_resources</code>	JSON	nullable	list[LabourResource]
<code>material_resources</code>	JSON	nullable	list[MaterialResource]
<code>tools_list</code>	JSON	nullable	Required tools
<code>special_equipment</code>	JSON	nullable	Special equipment needed
<code>ai_generated</code>	Boolean	default=False	AI-generated flag
<code>ai_confidence</code>	Float	nullable	AI confidence (0-1)
<code>status</code>	String(20)	default="DRAFT"	ApprovalStatus value

B.8 work_packages

Field	Type	Constraints	Description
work_package_id	String(50)	PK , default=uuid	WP identifier
name	String(40)	NOT NULL	WP name (ALL CAPS, max 40)
code	String(50)	indexed	WP code
node_id	String(50)	FK hierarchy_nodes.node_id	Equipment node
frequency_value	Float	NOT NULL	Interval value
frequency_unit	String(30)	NOT NULL	FrequencyUnit value
constraint	String(20)	NOT NULL	ONLINE or OFFLINE
access_time_hours	Float	default=0.0	Access/setup time
work_package_type	String(20)	NOT NULL	STANDALONE, SUPPRESSIVE, SEQUENTIAL
job_preparation	Text	nullable	Pre-job preparation notes
post_shutdown	Text	nullable	Post-shutdown activities
allocated_tasks	JSON	nullable	list[AllocatedTask]
labour_summary	JSON	nullable	LabourSummary
material_summary	JSON	nullable	list[MaterialSummaryEntry]
sap_upload_ref	JSON	nullable	SAPUploadRef
status	String(20)	default="DRAFT"	WPAApprovalStatus value

B.9 sap_upload_packages

Field	Type	Constraints	Description
package_id	String(50)	PK , default=uuid	Package identifier
generated_at	DateTime	default=now	Generation timestamp
plant_code	String(50)	NOT NULL	SAP plant code
maintenance_plan	JSON	nullable	SAPMaintenancePlan
maintenance_items	JSON	nullable	list[SAPMaintenanceItem]
task_lists	JSON	nullable	list[SAPTaskList]
status	String(20)	default="GENERATED"	SAPUploadStatus value

B.10 work_orders

Field	Type	Constraints	Description
work_order_id	String(50)	PK	WO identifier
order_type	String(10)	NOT NULL	PM01, PM02, PM03
equipment_id	String(100)	indexed	Equipment reference
equipment_tag	String(100)	default=""	Equipment TAG
priority	String(10)	NOT NULL	Priority (1-4)
status	String(20)	NOT NULL	WOHistoryStatus value
created_date	Date	NOT NULL	Creation date
actual_duration_hours	Float	nullable	Actual work duration
description	Text	default=""	WO description
materials_consumed	JSON	nullable	list[MaterialConsumed]

B.11 health_scores

Field	Type	Constraints	Description
score_id	String(50)	PK , default=uuid	Score identifier
node_id	String(50)	FK hierarchy_nodes.node_id	Assessed node
plant_id	String(50)	NOT NULL	Plant reference
equipment_tag	String(100)	NOT NULL	Equipment TAG
calculated_at	DateTime	default=now	Calculation timestamp
dimensions	JSON	nullable	list[HealthScoreDimension]
composite_score	Float	default=0.0	Weighted composite (0-100)
health_class	String(20)	default="UNKNOWN"	HEALTHY, AT_RISK, CRITICAL, UNKNOWN
trend	String(20)	default=""	IMPROVING, STABLE, DEGRADING
recommendations	JSON	nullable	Action recommendations

B.12 `kpi_metrics`

Field	Type	Constraints	Description
<code>metrics_id</code>	String(50)	PK , default=uuid	Metrics identifier
<code>plant_id</code>	String(50)	NOT NULL	Plant reference
<code>equipment_id</code>	String(100)	nullable	Equipment reference (null = plant-level)
<code>period_start</code>	Date	NOT NULL	Period start
<code>period_end</code>	Date	NOT NULL	Period end
<code>calculated_at</code>	DateTime	default=now	Calculation timestamp
<code>mtbf_days</code>	Float	nullable	Mean Time Between Failures (days)
<code>mttr_hours</code>	Float	nullable	Mean Time To Repair (hours)
<code>availability_pct</code>	Float	nullable	Availability (0-100%)
<code>oeo_pct</code>	Float	nullable	Overall Equipment Effectiveness (0-100%)
<code>schedule_compliance_pct</code>	Float	nullable	Schedule compliance (0-100%)
<code>backlog_hours</code>	Float	default=0.0	Pending backlog hours
<code>pm_compliance_pct</code>	Float	nullable	PM compliance (0-100%)
<code>total_work_orders</code>	Integer	default=0	Total WO count
<code>corrective_wo_count</code>	Integer	default=0	Corrective WO count
<code>preventive_wo_count</code>	Integer	default=0	Preventive WO count
<code>reactive_ratio_pct</code>	Float	nullable	Corrective / Total ratio (0-100%)

B.13 `failure_predictions`

Field	Type	Constraints	Description
<code>prediction_id</code>	String(50)	PK , default=uuid	Prediction identifier
<code>equipment_id</code>	String(100)	indexed	Equipment reference
<code>equipment_tag</code>	String(100)	NOT NULL	Equipment TAG
<code>predicted_at</code>	DateTime	default=now	Prediction timestamp
<code>weibull_params</code>	JSON	nullable	WeibullParameters (beta, eta, gamma, r_squared)
<code>current_age_days</code>	Float	NOT NULL	Current age in days
<code>reliability_current</code>	Float	NOT NULL	R(t) at current age (0-1)
<code>predicted_failure_window_days</code>	Float	NOT NULL	Days until predicted failure
<code>confidence_level</code>	Float	default=0.9	Confidence level (0.5-0.99)
<code>risk_score</code>	Float	default=0.0	Risk score (0-100)
<code>failure_pattern</code>	String(30)	nullable	FailurePattern value
<code>recommendation</code>	Text	default=""	Action recommendation
<code>status</code>	String(20)	default="DRAFT"	ApprovalStatus value

B.14 capa_items

Field	Type	Constraints	Description
capa_id	String(50)	PK , default=uuid	CAPA identifier
capa_type	String(20)	NOT NULL	CORRECTIVE or PREVENTIVE
title	String(300)	NOT NULL	CAPA title
description	Text	NOT NULL	CAPA description
plant_id	String(50)	NOT NULL	Plant reference
equipment_id	String(100)	nullable	Equipment reference
source	String(100)	NOT NULL	Origin (audit, incident, etc.)
root_cause	Text	nullable	Root cause description
current_phase	String(10)	default="PLAN"	PDCAPhase value
status	String(20)	default="OPEN"	CAPAStatus value
assigned_to	String(100)	default=""	Responsible person
created_at	DateTime	default=now	Creation timestamp
target_date	Date	nullable	Due date
closed_at	DateTime	nullable	Closure timestamp
verified_at	DateTime	nullable	Verification timestamp
actions_planned	JSON	nullable	Planned action items
actions_completed	JSON	nullable	Completed action items
effectiveness_verified	Boolean	default=False	Verification flag
lessons_learned	Text	nullable	Lessons learned text

B.15 variance_alerts

Field	Type	Constraints	Description
alert_id	String(50)	PK , default=uuid	Alert identifier
plant_id	String(50)	NOT NULL	Alerting plant
plant_name	String(200)	NOT NULL	Plant name
metric_name	String(100)	NOT NULL	KPI metric name
plant_value	Float	NOT NULL	Plant metric value
portfolio_mean	Float	NOT NULL	Portfolio average
portfolio_std	Float	NOT NULL	Portfolio standard deviation
z_score	Float	NOT NULL	Z-score deviation
variance_level	String(20)	NOT NULL	NORMAL, WARNING, CRITICAL
detected_at	DateTime	default=now	Detection timestamp
message	Text	default=""	Alert message

B.16 expert_cards

Field	Type	Constraints	Description
expert_id	String(50)	PK , default=uuid	Expert card identifier
user_id	String(100)	indexed	User account ID
name	String(200)	NOT NULL	Expert name
role	String(50)	NOT NULL	StakeholderRole value
plant_id	String(50)	NOT NULL	Home plant
domains	JSON	nullable	list[ExpertDomain]
equipment_expertise	JSON	nullable	Equipment TAGs known
certifications	JSON	nullable	Certification list
years_experience	Integer	default=0	Years of experience
resolution_count	Integer	default=0	Issues resolved count
last_active	DateTime	nullable	Last activity timestamp
contact_method	String(200)	default=""	Preferred contact
languages	JSON	nullable	Spoken languages

B.17 audit_log

Field	Type	Constraints	Description
id	Integer	PK , autoincrement	Log entry ID
entity_type	String(50)	indexed	Entity type name
entity_id	String(50)	nullable	Entity identifier
action	String(20)	NOT NULL	CREATE, UPDATE, DELETE, APPROVE
payload	JSON	nullable	Change payload
user	String(100)	default="system"	Acting user
timestamp	DateTime	default=now, indexed	Event timestamp

B.18 field_captures

Field	Type	Constraints	Description
capture_id	String(50)	PK , default=uuid	Capture identifier
technician_id	String(100)	NOT NULL	Technician user ID
capture_type	String(20)	NOT NULL	CaptureType value
language	String(5)	default="en"	Detected language
raw_text	Text	nullable	Text input content
raw_voice_text	Text	nullable	Voice transcription
images	JSON	nullable	list[CaptureImage]
equipment_tag_manual	String(100)	nullable	Manually-entered TAG
location_hint	String(200)	nullable	Location description
created_at	DateTime	default=now	Capture timestamp

B.19 work_requests

Field	Type	Constraints	Description
request_id	String(50)	PK , default=uuid	Request identifier
source_capture_id	String(50)	FK field_captures.capture_id, nullable	Source capture
status	String(30)	default="DRAFT"	WorkRequestStatus value
equipment_id	String(100)	NOT NULL	Equipment reference
equipment_tag	String(100)	NOT NULL	Equipment TAG
equipment_confidence	Float	default=0.0	ID confidence (0-1)
resolution_method	String(30)	default="MANUAL"	ResolutionMethod value
problem_description	JSON	nullable	ProblemDescription
ai_classification	JSON	nullable	AIClassification
spare_parts	JSON	nullable	list[SuggestedSparePart]
image_analysis	JSON	nullable	ImageAnalysis
validation	JSON	nullable	Validation
created_at	DateTime	default=now	Creation timestamp

Indexes: ix_work_requests_status

B.20 planner_recommendations

Field	Type	Constraints	Description
recommendation_id	String(50)	PK , default=uuid	Recommendation identifier
work_request_id	String(50)	FK work_requests.request_id	Linked work request
resource_analysis	JSON	nullable	ResourceAnalysis
scheduling_suggestion	JSON	nullable	SchedulingSuggestion
risk_assessment	JSON	nullable	RiskAssessment
planner_action	String(20)	NOT NULL	PlannerAction value
ai_confidence	Float	default=0.0	AI confidence (0-1)
generated_at	DateTime	default=now	Generation timestamp

B.21 backlog_items

Field	Type	Constraints	Description
backlog_id	String(50)	PK , default=uuid	Backlog item identifier
work_request_id	String(50)	FK work_requests.request_id, nullable	Source work request
equipment_id	String(100)	NOT NULL	Equipment reference
equipment_tag	String(100)	NOT NULL	Equipment TAG
priority	String(20)	NOT NULL	Priority value
wo_type	String(10)	NOT NULL	PM01, PM02, PM03
status	String(30)	default="AWAITING_APPROVAL"	BacklogStatus value
blocking_reason	String(200)	nullable	Reason for block
estimated_hours	Float	default=4.0	Estimated duration
specialties	JSON	nullable	Required specialties
materials_ready	Boolean	default=False	Materials available flag
shutdown_required	Boolean	default=False	Shutdown needed flag
age_days	Integer	default=0	Age in backlog
created_at	DateTime	default=now	Creation timestamp

Indexes: ix_backlog_items_priority, ix_backlog_items_status

B.22 optimized_backlogs

Field	Type	Constraints	Description
optimization_id	String(50)	PK , default=uuid	Optimization identifier
plant_id	String(50)	NOT NULL	Plant reference
period_start	Date	NOT NULL	Period start
period_end	Date	NOT NULL	Period end
total_items	Integer	default=0	Total backlog items
stratification	JSON	nullable	BacklogStratification
work_packages	JSON	nullable	list[BacklogWorkPackage]
schedule	JSON	nullable	list[ScheduleEntry]
alerts	JSON	nullable	list[BacklogAlert]
status	String(20)	default="DRAFT"	Status
generated_at	DateTime	default=now	Generation timestamp

B.23 workforce

Field	Type	Constraints	Description
worker_id	String(50)	PK , default=uuid	Worker identifier
name	String(200)	NOT NULL	Worker name
specialty	String(50)	NOT NULL	LabourSpecialty value
shift	String(20)	NOT NULL	ShiftType value
plant_id	String(50)	NOT NULL	Plant reference
available	Boolean	default=True	Availability flag
certifications	JSON	nullable	Certifications list

B.24 shutdown_calendar

Field	Type	Constraints	Description
shutdown_id	String(50)	PK , default=uuid	Shutdown identifier
plant_id	String(50)	NOT NULL	Plant reference
start_date	Date	NOT NULL	Start date
end_date	Date	NOT NULL	End date
shutdown_type	String(30)	NOT NULL	MINOR_8H or MAJOR_20H_PLUS
areas	JSON	nullable	Affected areas
description	Text	default=""	Description

B.25 inventory_items

Field	Type	Constraints	Description
item_id	String(50)	PK , default=uuid	Item identifier
material_code	String(50)	indexed	SAP material code
warehouse_id	String(50)	default="WH-01"	Warehouse ID
description	String(300)	NOT NULL	Part description
quantity_on_hand	Integer	default=0	On-hand quantity
quantity_reserved	Integer	default=0	Reserved quantity
quantity_available	Integer	default=0	Available quantity
min_stock	Integer	default=0	Minimum stock level
reorder_point	Integer	default=0	Reorder trigger point
last_movement_date	Date	nullable	Last stock movement

B.26 weekly_programs

Field	Type	Constraints	Description
program_id	String(50)	PK , default=uuid	Program identifier
plant_id	String(50)	NOT NULL	Plant reference
week_number	Integer	NOT NULL	ISO week number
year	Integer	NOT NULL	Year
status	String(20)	default="DRAFT"	WeeklyProgramStatus value
work_packages	JSON	nullable	Scheduled work packages
total_hours	Float	default=0.0	Total planned hours
resource_slots	JSON	nullable	list[ResourceSlot]
conflicts	JSON	nullable	list[ResourceConflict]
support_tasks	JSON	nullable	list[SupportTask]
created_at	DateTime	default=now	Creation timestamp
finalized_at	DateTime	nullable	Finalization timestamp

Indexes: ix_weekly_programs_plant_year_week

B.27 shutdown_events

Field	Type	Constraints	Description
shutdown_id	String(50)	PK , default=uuid	Shutdown identifier
plant_id	String(50)	NOT NULL	Plant reference
name	String(200)	NOT NULL	Shutdown name
status	String(20)	default="PLANNED"	ShutdownStatus value
planned_start	DateTime	NOT NULL	Planned start datetime
planned_end	DateTime	NOT NULL	Planned end datetime
actual_start	DateTime	nullable	Actual start
actual_end	DateTime	nullable	Actual end
planned_hours	Float	default=0.0	Planned duration
actual_hours	Float	default=0.0	Actual duration
work_orders	JSON	nullable	WO list
completed_work_orders	JSON	nullable	Completed WO list
completion_pct	Float	default=0.0	Completion percentage
delay_hours	Float	default=0.0	Total delay
delay_reasons	JSON	nullable	Delay reason list
created_at	DateTime	default=now	Creation timestamp

B.28 moc_requests

Field	Type	Constraints	Description
<code>moc_id</code>	String(50)	PK , default=uuid	MoC identifier
<code>plant_id</code>	String(50)	NOT NULL	Plant reference
<code>title</code>	String(300)	NOT NULL	MoC title
<code>description</code>	Text	default=""	MoC description
<code>category</code>	String(50)	default="EQUIPMENT_MODIFICATION"	MoCCategory value
<code>status</code>	String(20)	default="DRAFT"	MoCStatus value
<code>risk_level</code>	String(20)	default="LOW"	RiskLevel value
<code>requester_id</code>	String(50)	default=""	Requester ID
<code>reviewer_id</code>	String(50)	default=""	Reviewer ID
<code>approver_id</code>	String(50)	default=""	Approver ID
<code>affected_equipment</code>	JSON	nullable	Affected equipment list
<code>affected_procedures</code>	JSON	nullable	Affected procedures
<code>risk_assessment</code>	Text	default=""	Risk assessment text
<code>impact_analysis</code>	Text	default=""	Impact analysis text
<code>created_at</code>	DateTime	default=now	Creation timestamp
<code>submitted_at</code>	DateTime	nullable	Submission timestamp
<code>approved_at</code>	DateTime	nullable	Approval timestamp
<code>closed_at</code>	DateTime	nullable	Closure timestamp

B.29 spare_part_analyses

Field	Type	Constraints	Description
<code>analysis_id</code>	String(50)	PK , default=uuid	Analysis identifier
<code>plant_id</code>	String(50)	NOT NULL	Plant reference
<code>analyzed_at</code>	DateTime	default=now	Analysis timestamp
<code>total_parts</code>	Integer	default=0	Total parts analyzed
<code>results</code>	JSON	nullable	list[SparePartAnalysis]
<code>total_inventory_value</code>	Float	default=0.0	Total inventory value
<code>recommended_reduction_pct</code>	Float	default=0.0	Recommended stock reduction

B.30 rbi_assessments

Field	Type	Constraints	Description
assessment_id	String(50)	PK , default=uuid	Assessment identifier
plant_id	String(50)	NOT NULL	Plant reference
analyzed_at	DateTime	default=now	Assessment timestamp
total_equipment	Integer	default=0	Equipment count assessed
assessments	JSON	nullable	list[RBIAssessment]
high_risk_count	Integer	default=0	High-risk count
overdue_count	Integer	default=0	Overdue inspections

B.31 reports

Field	Type	Constraints	Description
report_id	String(50)	PK , default=uuid	Report identifier
report_type	String(30)	NOT NULL	ReportType value
plant_id	String(50)	NOT NULL	Plant reference
period_start	DateTime	NOT NULL	Report period start
period_end	DateTime	NOT NULL	Report period end
generated_at	DateTime	default=now	Generation timestamp
content	JSON	nullable	Report content
metadata_json	JSON	nullable	Report metadata

Indexes: ix_reports_plant_type

B.32 notifications

Field	Type	Constraints	Description
notification_id	String(50)	PK , default=uuid	Notification identifier
notification_type	String(30)	NOT NULL	NotificationType value
level	String(20)	default="INFO"	NotificationLevel value
plant_id	String(50)	NOT NULL	Plant reference
equipment_id	String(100)	nullable	Equipment reference
title	String(300)	NOT NULL	Notification title
message	Text	default=""	Notification body
acknowledged	Boolean	default=False	Acknowledged flag
created_at	DateTime	default=now	Creation timestamp
acknowledged_at	DateTime	nullable	Acknowledgement timestamp

B.33 rca_analyses

Field	Type	Constraints	Description
analysis_id	String(50)	PK , default=uuid	Analysis identifier
event_description	Text	NOT NULL	Event description
plant_id	String(50)	NOT NULL	Plant reference
equipment_id	String(100)	nullable	Equipment reference
level	String(10)	default="LEVEL_1"	RCA level (1/2/3)
status	String(30)	default="OPEN"	RCAStatus value
team_members	JSON	nullable	Team member list
analysis_5w2h	JSON	nullable	Analysis5W2H data
cause_effect	JSON	nullable	CauseEffectDiagram
evidence_5p	JSON	nullable	list[Evidence5P]
solutions	JSON	nullable	list[Solution]
created_at	DateTime	default=now	Creation timestamp
completed_at	DateTime	nullable	Completion timestamp

Indexes: ix_rca_analyses_plant_status

B.34 planning_kpi_snapshots

Field	Type	Constraints	Description
snapshot_id	String(50)	PK , default=uuid	Snapshot identifier
plant_id	String(50)	NOT NULL	Plant reference
period_start	Date	NOT NULL	Period start
period_end	Date	NOT NULL	Period end
kpis	JSON	nullable	list[PlanningKPIValue]
overall_health	String(20)	default="UNKNOWN"	HEALTHY, AT_RISK, CRITICAL
on_target_count	Integer	default=0	KPIs on target
below_target_count	Integer	default=0	KPIs below target
calculated_at	DateTime	default=now	Calculation timestamp

Indexes: ix_planning_kpi_plant_period

B.35 de_kpi_snapshots

Field	Type	Constraints	Description
snapshot_id	String(50)	PK , default=uuid	Snapshot identifier
plant_id	String(50)	NOT NULL	Plant reference
period_start	Date	NOT NULL	Period start
period_end	Date	NOT NULL	Period end
kpis	JSON	nullable	list[DEKPIValue]
overall_compliance	Float	default=0.0	Overall compliance score
program_score	Float	default=0.0	DE program score
maturity_level	String(20)	default="INITIAL"	INITIAL, DEVELOPING, ESTABLISHED, OPTIMIZING
calculated_at	DateTime	default=now	Calculation timestamp

Indexes: ix_de_kpi_plant_period

B.36 user_feedback

Field	Type	Constraints	Description
feedback_id	String(50)	PK , default=uuid	Feedback identifier
page	String(100)	NOT NULL	Page/feature name
rating	Integer	NOT NULL	Rating (1-5)
comment	Text	default=""	Free-text comment
created_at	DateTime	default=now	Submission timestamp

B.37 fmeca_worksheets

Field	Type	Constraints	Description
worksheet_id	String(50)	PK , default=uuid	Worksheet identifier
equipment_id	String(100)	indexed	Equipment reference
equipment_tag	String(100)	default=""	Equipment TAG
equipment_name	String(300)	default=""	Equipment name
status	String(20)	default="DRAFT"	FMECAWorksheetStatus value
current_stage	String(30)	default="STAGE_1_FUNCTIONS"	FMECAStage value
rows	JSON	nullable	Worksheet row data
stage_completion	JSON	nullable	Per-stage completion status
analyst	String(100)	default=""	Analyst name
created_at	DateTime	default=now	Creation timestamp
completed_at	DateTime	nullable	Completion timestamp

Section C: API Endpoints

All router endpoints from `api/routers/`.

C.1 Hierarchy (`/hierarchy`)

Method	Path	Request Fields	Response Fields
GET	<code>/hierarchy/plants</code>	--	plant_id, name, name_fr, location
POST	<code>/hierarchy/plants</code>	plant_id, name, name_fr, name_ar, location	plant_id, name
GET	<code>/hierarchy/nodes</code>	?plant_id, ?node_type, ?parent_node_id	node_id, node_type, name, name_fr, code, parent_node_id, level, plant_id, tag, criticality, status, metadata
POST	<code>/hierarchy/nodes</code>	node_type, name, code, level, plant_id, parent_node_id, tag, status, metadata	(same as GET node)
GET	<code>/hierarchy/nodes/{node_id}</code>	--	node_id, node_type, name, name_fr, code, parent_node_id, level, plant_id, tag, criticality, status, metadata
GET	<code>/hierarchy/nodes/{node_id}/tree</code>	--	list of node objects
GET	<code>/hierarchy/stats</code>	?plant_id	dict of node_type counts

C.2 Criticality (`/criticality`)

Method	Path	Request Fields	Response Fields
POST	<code>/criticality/assess</code>	node_id, criteria_scores, probability, ?method, ?assessed_by	assessment_id, node_id, overall_score, risk_class, criteria_scores, probability, status
GET	<code>/criticality/{node_id}</code>	--	assessment_id, node_id, overall_score, risk_class, criteria_scores, probability, status
PUT	<code>/criticality/{assessment_id}/approve</code>	--	assessment_id, status
POST	<code>/criticality/risk-class</code>	overall_score	risk_class result

C.3 FMEA (/fmea)

Method	Path	Request Fields	Response Fields
POST	/fmea/failure-modes	functional_failure_id, what, mechanism, cause, failure_consequence, strategy_type, ...	failure_mode_id, what, mechanism, cause, strategy_type
GET	/fmea/failure-modes/{fm_id}	--	failure_mode_id, what, mechanism, cause, failure_consequence, strategy_type, failure_effect
GET	/fmea/failure-modes	?functional_failure_id	list: failure_mode_id, what, mechanism, cause, strategy_type
POST	/fmea/validate-combination	mechanism, cause	validation result
GET	/fmea/fm-combinations	?mechanism	valid combinations
POST	/fmea/rcm-decide	(RCM decision inputs)	strategy recommendation
POST	/fmea/functions	node_id, function_type, description, description_fr	function_id, node_id, function_type
POST	/fmea/functional-failures	function_id, failure_type, description, description_fr	failure_id, function_id, failure_type
POST	/fmea/fmeca/worksheets	equipment_id, equipment_tag, equipment_name, analyst	worksheet data
GET	/fmea/fmeca/worksheets/{worksheet_id}	--	full worksheet data
POST	/fmea/fmeca/rpn	severity, occurrence, detection	rpn_value, rpn_category
PUT	/fmea/fmeca/worksheets/{worksheet_id}/run-decisions	--	updated worksheet with RCM decisions
GET	/fmea/fmeca/worksheets/{worksheet_id}/summary	--	summary statistics

C.4 Tasks (/tasks)

Method	Path	Request Fields	Response Fields
POST	/tasks/	name, task_type, constraint, frequency_value, frequency_unit, ...	task_id, name, task_type, status
GET	/tasks/{task_id}	--	task_id, name, task_type, constraint, frequency_value, frequency_unit, status, labour_resources, material_resources
GET	/tasks/	?failure_mode_id, ?status	list: task_id, name, task_type, status
POST	/tasks/link-fm/{task_id}/{fm_id}	--	task_id, failure_mode_id
POST	/tasks/validate-name	name, ?task_type	validation result
POST	/tasks/validate-wp-name	name	validation result

C.5 Work Packages (/work-packages)

Method	Path	Request Fields	Response Fields
POST	/work-packages/	name, code, node_id, frequency_value, frequency_unit, constraint, work_package_type	work_package_id, name, code, status
GET	/work-packages/{wp_id}	--	work_package_id, name, code, node_id, frequency_value, frequency_unit, constraint, work_package_type, status, allocated_tasks, labour_summary
GET	/work-packages/	?node_id, ?status	list: work_package_id, name, code, status
PUT	/work-packages/{wp_id}/approve	--	work_package_id, status
POST	/work-packages/group	items (list of tasks)	grouped work packages
POST	/work-packages/{wp_id}/work-instruction	equipment_name, equipment_tag, tasks	work instruction document

C.6 SAP (/sap)

Method	Path	Request Fields	Response Fields
POST	/sap/generate-upload	plant_code, maintenance_plan, maintenance_items, task_lists	package_id, status
GET	/sap/uploads/{package_id}	--	package_id, plant_code, status, generated_at, maintenance_plan, maintenance_items, task_lists
GET	/sap/uploads	?plant_code	list: package_id, plant_code, status
PUT	/sap/uploads/{package_id}/approve	--	approval result
POST	/sap/validate-transition	entity_type, current_state, target_state	validation result
GET	/sap/mock/{transaction}	--	mock SAP data

C.7 Analytics (/analytics)

Method	Path	Request Fields	Response Fields
POST	/analytics/health-score	node_id, plant_id, equipment_tag, risk_class, ? pending_backlog_hours, ?capacity_hours_per_week, ? total_failure_modes, ?fm_with_strategy, ?active_alerts, ?critical_alerts, ?planned_wo, ?executed_on_time	AssetHealthScore
POST	/analytics/kpis	plant_id, ?failure_dates, ?total_period_hours, ?total_downtime_hours	KPIMetrics
POST	/analytics/weibull-fit	failure_intervals	WeibullParameters
POST	/analytics/weibull-predict	equipment_id, equipment_tag, failure_intervals, current_age_days, ? confidence_level	FailurePrediction
POST	/analytics/variance-detect	snapshots	list of PlantVarianceAlert
GET	/analytics/variance-alerts	--	list: alert_id, plant_id, metric_name, z_score, variance_level

C.8 Capture (/capture)

Method	Path	Request Fields	Response Fields
POST	/capture/	technician_id, capture_type, language, raw_text, raw_voice_text, images, equipment_tag_manual, location_hint	capture result with work request
GET	/capture/	--	list: capture_id, technician_id, capture_type, language, equipment_tag_manual, created_at
GET	/capture/{capture_id}	--	capture_id, technician_id, capture_type, language, raw_text, raw_voice_text, equipment_tag_manual, location_hint, created_at

C.9 Work Requests (/work-requests)

Method	Path	Request Fields	Response Fields
GET	/work-requests/	?status	list: request_id, status, equipment_tag, equipment_confidence, ai_classification, created_at
GET	/work-requests/{request_id}	--	request_id, source_capture_id, status, equipment_id, equipment_tag, equipment_confidence, problem_description, ai_classification, spare_parts, image_analysis, validation, created_at
PUT	/work-requests/{request_id}/validate	action (APPROVE/REJECT/MODIFY), ? modifications	validation result
POST	/work-requests/{request_id}/classify	--	classification result

C.10 Planner (/planner)

Method	Path	Request Fields	Response Fields
POST	/planner/{work_request_id}/recommend	--	recommendation result
GET	/planner/recommendations/{recommendation_id}	--	recommendation_id, work_request_id, planner_action, ai_confidence, resource_analysis, scheduling_suggestion, risk_assessment, generated_at
PUT	/planner/recommendations/{recommendation_id}/action	action (APPROVE/MODIFY/ESCALATE/DEFER), ?modifications	action result

C.11 Backlog (/backlog)

Method	Path	Request Fields	Response Fields
GET	/backlog/	?status, ?priority, ?equipment_tag	backlog items list
POST	/backlog/add/{work_request_id}	--	added backlog item
POST	/backlog/optimize	?plant_id, ?period_days	optimization result
GET	/backlog/optimizations/{optimization_id}	--	optimization_id, plant_id, status, total_items, stratification, work_packages, schedule, alerts, generated_at
PUT	/backlog/optimizations/{optimization_id}/approve	--	approval result
GET	/backlog/schedule	--	current schedule

C.12 Scheduling (/scheduling)

Method	Path	Request Fields	Response Fields
POST	/scheduling/programs	?plant_id, ?week_number, ?year	program data
GET	/scheduling/programs	?plant_id, ?status	list of programs
GET	/scheduling/programs/{program_id}	--	program_id, plant_id, week_number, year, status, total_hours, work_packages, resource_slots, conflicts, support_tasks, created_at, finalized_at
PUT	/scheduling/programs/{program_id}/finalize	--	finalization result
PUT	/scheduling/programs/{program_id}/activate	--	activation result
PUT	/scheduling/programs/{program_id}/complete	--	completion result
GET	/scheduling/programs/{program_id}/gantt	--	Gantt chart data
GET	/scheduling/programs/{program_id}/gantt/export	--	Excel file download

C.13 Reliability (/reliability)

Method	Path	Request Fields	Response Fields
POST	/reliability/spare-parts/analyze	?plant_id, parts	SparePartOptimizationResult
POST	/reliability/shutdowns	plant_id, name, planned_start, planned_end, work_orders	shutdown data
GET	/reliability/shutdowns/{shutdown_id}	--	full shutdown data
PUT	/reliability/shutdowns/{shutdown_id}/start	--	updated shutdown
PUT	/reliability/shutdowns/{shutdown_id}/complete	--	completed shutdown
POST	/reliability/moc	plant_id, title, description, category, requester_id, affected_equipment, risk_level	MoC data
GET	/reliability/moc	?plant_id, ?status	list of MoCs
GET	/reliability/moc/{moc_id}	--	full MoC data
PUT	/reliability/moc/{moc_id}/advance	action, ...	updated MoC
POST	/reliability/ocr/analyze	equipment_id, failure_rate, mttr_hours, cost_per_failure, cost_per_pm, current_pm_interval_days	OCRAAnalysisResult
POST	/reliability/jackknife/analyze	?plant_id, equipment_data	JackKnifeResult
POST	/reliability/pareto/analyze	?plant_id, ?metric_type, records	ParetoResult
POST	/reliability/lcc/calculate	equipment_id, acquisition_cost, installation_cost, annual_operating_cost, annual_maintenance_cost, expected_life_years, discount_rate, salvage_value	LCCResult
POST	/reliability/rbi/assess	?plant_id, equipment_list	RBIResult

C.14 Reporting (/reporting)

Method	Path	Request Fields	Response Fields
POST	/reporting/reports/weekly	?plant_id, week/week_number, year, ...	WeeklyReport
POST	/reporting/reports/monthly	?plant_id, month, year, ...	MonthlyKPIReport
POST	/reporting/reports/quarterly	?plant_id, quarter, year, ...	QuarterlyReviewReport
GET	/reporting/reports	?plant_id, ?report_type	list of reports
GET	/reporting/reports/{report_id}	--	full report data
POST	/reporting/de-kpis/calculate	(DEKPIInput fields)	DEKPIs
POST	/reporting/de-kpis/program-health	(DE KPI data)	DEProgramHealth
POST	/reporting/notifications/generate	plant_id, config	NotificationResult
GET	/reporting/notifications	?plant_id, ?level	list of notifications
PUT	/reporting/notifications/{notification_id}/ack	--	acknowledgement result
POST	/reporting/import/validate	source, rows	ImportResult
POST	/reporting/export	export_type, ...	ExportResult
POST	/reporting/cross-module/analyze	plant_id, ...	CrossModuleSummary

C.15 RCA (/rca)

Method	Path	Request Fields	Response Fields
POST	/rca/analyses	event_description, ?plant_id, ?equipment_id, ?max_consequence, ?frequency, ?team_members	RCA analysis data
GET	/rca/analyses	?plant_id, ?status	list of RCA analyses
GET	/rca/analyses/summary	?plant_id	summary statistics
GET	/rca/analyses/{analysis_id}	--	full RCA analysis
POST	/rca/analyses/{analysis_id}/5w2h	what, when, where, who, why, how, how_much	5W+2H result
PUT	/rca/analyses/{analysis_id}/advance	status	status transition result
POST	/rca/planning-kpis/calculate	(PlanningKPIInput fields)	PlanningKPIs
GET	/rca/planning-kpis	?plant_id	list of snapshots
POST	/rca/de-kpis/calculate	(DEKPIInput fields)	DEKPIs
GET	/rca/de-kpis	?plant_id	list of DE KPI snapshots

C.16 Dashboard (/dashboard)

Method	Path	Request Fields	Response Fields
GET	/dashboard/executive/{plant_id}	--	plant_id, total_reports, recent_reports, total_notifications, critical_alerts, recent_notifications
GET	/dashboard/kpi-summary/{plant_id}	--	plant_id, has_data, report
GET	/dashboard/alerts/{plant_id}	--	plant_id, total_active, alerts

C.17 Admin (/admin)

Method	Path	Request Fields	Response Fields
POST	/admin/seed-database	--	seed result
GET	/admin/audit-log	?entity_type, ?limit	list: id, entity_type, entity_id, action, user, timestamp
GET	/admin/stats	--	plants, hierarchy_nodes, total_nodes
DELETE	/admin/reset-database	--	status confirmation
GET	/admin/agent-status	--	agent status data
POST	/admin/feedback	page, rating, comment	feedback_id, status
GET	/admin/feedback	?page, ?limit	list: feedback_id, page, rating, comment, created_at

Section D: KPI Definitions

D.1 Planning KPIs (11 KPIs -- GFSN REF-14 S8)

Source: tools/engines/planning_kpi_engine.py

#	KPI Name	Formula	GFSN Target	Unit
1	WO Completion Rate	wo_completed / wo_planned * 100	>= 90%	%
2	Man-hour Compliance	manhours_actual / manhours_planned * 100	85--115%	%
3	PM Plan Compliance	pm_executed / pm_planned * 100	>= 95%	%
4	Backlog Weeks	backlog_hours / weekly_capacity_hours	<= 4 weeks	weeks
5	Reactive Work %	corrective_count / total_wo * 100	<= 20%	%
6	Schedule Adherence	schedule_compliance_executed / schedule_compliance_planned * 100	>= 85%	%
7	Release Horizon	release_horizon_days (direct input)	<= 7 days	days
8	Pending Notices %	pending_notices / total_notices * 100	<= 15%	%
9	Scheduled Capacity	scheduled_capacity_hours / total_capacity_hours * 100	80--95%	%
10	Proactive Work %	proactive_wo / total_wo * 100	>= 70%	%
11	Planning Efficiency	planned_wo / total_wo * 100	>= 85%	%

Overall Health Classification:

- = 9 KPIs on target: **HEALTHY**
- 6-8 KPIs on target: **AT_RISK**
- < 6 KPIs on target: **CRITICAL**

D.2 Defect Elimination KPIs (5 KPIs -- GFSN REF-15)

Source: `tools/engines/rca_engine.py` (compute_de_kpis), `tools/engines/de_kpi_engine.py`

#	KPI Name	Formula	Target	Unit
1	Event Reporting Compliance	<code>events_reported / events_required * 100</code>	>= 95%	%
2	Meeting Compliance	<code>meetings_held / meetings_required * 100</code>	>= 90%	%
3	Implementation Progress	<code>actions_implemented / actions_planned * 100</code>	>= 80%	%
4	Savings Effectiveness	<code>savings_achieved / savings_target * 100</code>	>= 70%	%
5	Frequency Reduction	<code>(failures_previous - failures_current) / failures_previous * 100</code>	>= 10%	%

DE Program Maturity Levels (from `DEKPIEngine.assess_program_health`):

- Score >= 80: **OPTIMIZING**
- Score >= 60: **ESTABLISHED**
- Score >= 40: **DEVELOPING**
- Score < 40: **INITIAL**

D.3 Core Reliability KPIs (7 KPIs -- ISO 55002 S9.1)

Source: `tools/engines/kpi_engine.py`

#	KPI Name	Formula	Description
1	MTBF	<code>sum(intervals between failure dates) / count(intervals)</code>	Mean Time Between Failures in days. Requires >= 2 failure events.
2	MTTR	<code>sum(repair_durations) / count(repairs)</code>	Mean Time To Repair in hours. Only positive durations counted.
3	Availability	<code>(total_period_hours - total_downtime_hours) / total_period_hours * 100</code>	Uptime percentage (0-100%).
4	OEE	<code>(availability/100) * (performance/100) * (quality/100) * 100</code>	Overall Equipment Effectiveness. Performance and quality default to 100% in MVP.
5	Schedule Compliance	<code>executed_on_time / planned_count * 100</code>	Percentage of WOs completed on schedule.
6	PM Compliance	<code>pm_executed / pm_planned * 100</code>	Preventive maintenance execution rate.
7	Reactive Ratio	<code>corrective_count / total_count * 100</code>	Percentage of work that is corrective/reactive.

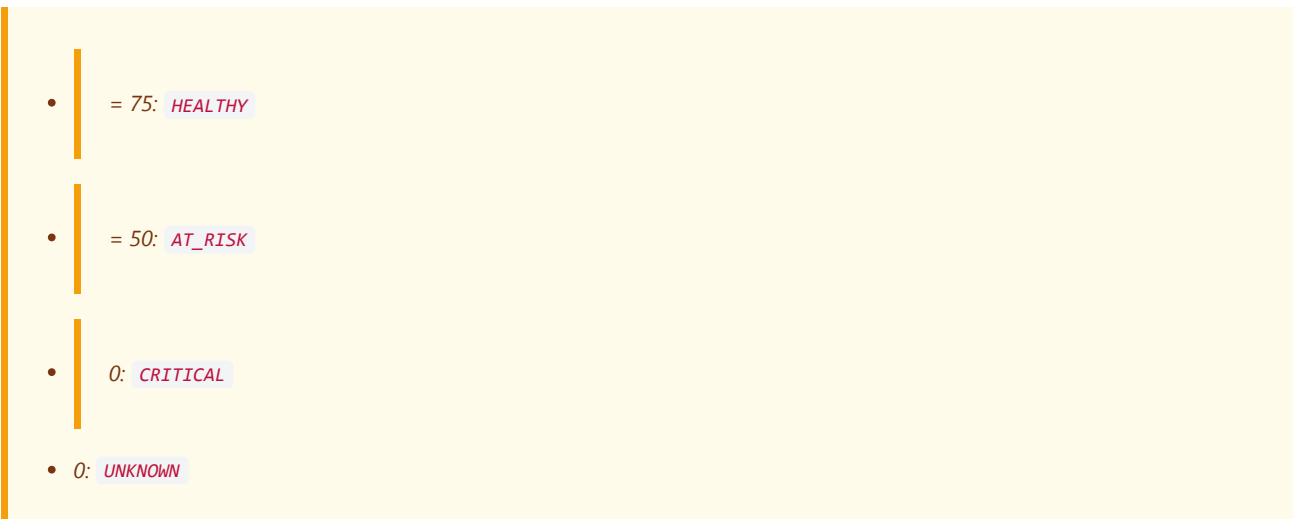
D.4 Asset Health Index (5 Dimensions -- REF-12 Rec 4)

Source: `tools/engines/health_score_engine.py`

#	Dimension	Default Weight	Score Calculation	Details
1	Criticality	0.25	Risk class mapping: I_LOW=90, II_MEDIUM=70, III_HIGH=40, IV_CRITICAL=15	Inverse: higher criticality = lower health score
2	Backlog Pressure	0.20	<code>100 * (1 - weeks_of_backlog / 8)</code> where weeks = pending_hours / capacity_per_week	0 pending = 100 (healthy); >= 8 weeks backlog = 0
3	Strategy Coverage	0.25	<code>100 * fm_with_strategy / total_failure_modes</code>	Percentage of failure modes covered by approved strategy
4	Condition Status	0.15	<code>100 * (1 - (active_alerts + critical_alerts * 2) / 10)</code>	Critical alerts weighted 3x; max threshold = 10
5	Execution Compliance	0.15	<code>100 * executed_on_time / planned_wo</code>	On-time work order execution rate

Composite Score: `sum(dimension_score * weight) / sum(weights)`

Health Classification:



Trend Detection: Delta > 5 = **IMPROVING** ; Delta < -5 = **DEGRADING** ; else **STABLE**

Section E: 72 Valid Failure Mode Combinations

The authoritative lookup table from `VALID_FM_COMBINATIONS` in `tools/models/schemas.py`.

Source: SRC-09 -- "Failure Modes (Mechanism + Cause).xlsx".

Any `FailureMode` must use one of these 72 mechanism+cause pairs.

#	Mechanism	Cause
1	ARCS	BREAKDOWN_IN_INSULATION
2	BLOCKS	CONTAMINATION
3	BLOCKS	EXCESSIVE_PARTICLE_SIZE
4	BLOCKS	INSUFFICIENT_FLUID_VELOCITY
5	BREAKS_FRACTURE_SEPARATES	CYCLIC_LOADING
6	BREAKS_FRACTURE_SEPARATES	MECHANICAL_OVERLOAD
7	BREAKS_FRACTURE_SEPARATES	THERMAL_OVERLOAD
8	CORRODES	BIO_ORGANISMS
9	CORRODES	CHEMICAL_ATTACK
10	CORRODES	CORROSIVE_ENVIRONMENT
11	CORRODES	CREVICE
12	CORRODES	DISSIMILAR_METALS_CONTACT
13	CORRODES	EXPOSURE_TO_ATMOSPHERE
14	CORRODES	EXPOSURE_TO_HIGH_TEMP_CORROSIVE
15	CORRODES	EXPOSURE_TO_HIGH_TEMP
16	CORRODES	EXPOSURE_TO_LIQUID_METAL
17	CORRODES	POOR_ELECTRICAL_CONNECTIONS
18	CORRODES	POOR_ELECTRICAL_INSULATION
19	CRACKS	AGE
20	CRACKS	CYCLIC_LOADING
21	CRACKS	EXCESSIVE_TEMPERATURE
22	CRACKS	HIGH_TEMP_CORROSIVE
23	CRACKS	IMPACT_SHOCK_LOADING
24	CRACKS	THERMAL_STRESSES
25	DEGRADES	AGE
26	DEGRADES	CHEMICAL_ATTACK
27	DEGRADES	CHEMICAL_REACTION
28	DEGRADES	CONTAMINATION
29	DEGRADES	ELECTRICAL_ARCING
30	DEGRADES	ENTRAINED_AIR
31	DEGRADES	EXCESSIVE_TEMPERATURE
32	DEGRADES	RADIATION
33	DISTORTS	IMPACT_SHOCK_LOADING
34	DISTORTS	MECHANICAL_OVERLOAD
35	DISTORTS	OFF_CENTER_LOADING

#	Mechanism	Cause
36	DISTORTS	USE
37	DRIFTS	EXCESSIVE_TEMPERATURE
38	DRIFTS	IMPACT_SHOCK_LOADING
39	DRIFTS	STRAY_CURRENT
40	DRIFTS	UNEVEN_LOADING
41	DRIFTS	USE
42	EXPIRES	AGE
43	IMMOBILISED	CONTAMINATION
44	IMMOBILISED	LACK_OF_LUBRICATION
45	LOOSES_PRELOAD	CREEP
46	LOOSES_PRELOAD	EXCESSIVE_TEMPERATURE
47	LOOSES_PRELOAD	VIBRATION
48	OPEN_CIRCUIT	ELECTRICAL_OVERLOAD
49	OVERHEATS_MELTS	CONTAMINATION
50	OVERHEATS_MELTS	ELECTRICAL_OVERLOAD
51	OVERHEATS_MELTS	LACK_OF_LUBRICATION
52	OVERHEATS_MELTS	MECHANICAL_OVERLOAD
53	OVERHEATS_MELTS	RELATIVE_MOVEMENT
54	OVERHEATS_MELTS	RUBBING
55	SEVERS	ABRASION
56	SEVERS	IMPACT_SHOCK_LOADING
57	SEVERS	MECHANICAL_OVERLOAD
58	SHORT_CIRCUITS	BREAKDOWN_IN_INSULATION
59	SHORT_CIRCUITS	CONTAMINATION
60	THERMALLY_OVERLOADS	MECHANICAL_OVERLOAD
61	THERMALLY_OVERLOADS	OVERCURRENT
62	WASHES_OFF	EXCESSIVE_FLUID_VELOCITY
63	WASHES_OFF	USE
64	WEARS	BREAKDOWN_OF_LUBRICATION
65	WEARS	ENTRAINED_AIR
66	WEARS	EXCESSIVE_FLUID_VELOCITY
67	WEARS	IMPACT_SHOCK_LOADING
68	WEARS	LOW_PRESSURE
69	WEARS	LUBRICANT_CONTAMINATION
70	WEARS	MECHANICAL_OVERLOAD

#	Mechanism	Cause
71	WEARS	METAL_TO_METAL_CONTACT
72	WEARS	RELATIVE_MOVEMENT

Section F: Excel Template Schemas (14 Templates)

All `.xlsx` data-loading templates generated by `templates/generate_templates.py`. Each template includes OCP green headers, frozen panes, data validation dropdowns, and 3-5 realistic phosphate example rows.

F.1 Template 01: `01_equipment_hierarchy.xlsx`

Plant and equipment hierarchy nodes (18 fields).

Field	Type	Validation	Description
<code>plant_id</code>	String	required	Plant identifier (e.g., OCP-JFC1)
<code>plant_name</code>	String	required	Plant name
<code>area_code</code>	String	required	Area code (e.g., BRY)
<code>area_name</code>	String	required	Area name
<code>system_code</code>	String	required	System code
<code>system_name</code>	String	required	System name
<code>equipment_tag</code>	String	required	Equipment tag (e.g., BRY-SAG-ML-001)
<code>equipment_description</code>	String	required	Equipment description
<code>equipment_type</code>	String	dropdown: EquipmentCategory	Equipment type
<code>manufacturer</code>	String	-	OEM manufacturer
<code>model</code>	String	-	Model number
<code>serial_number</code>	String	-	Serial number
<code>power_kw</code>	Float	-	Rated power (kW)
<code>weight_kg</code>	Float	-	Weight (kg)
<code>criticality</code>	String	dropdown: EquipmentCriticality	Criticality grade
<code>status</code>	String	dropdown: EquipmentStatus	Equipment status
<code>sap_func_loc</code>	String	-	SAP functional location
<code>installation_date</code>	Date	YYYY-MM-DD	Installation date

F.2 Template 02: `02_criticality_assessment.xlsx`

Criticality assessment scores (14 fields).

Field	Type	Validation	Description
equipment_tag	String	required	Equipment tag reference
method	String	dropdown: CriticalityMethod	R8 or GFSN method
safety	Integer	1-5	Safety consequence score
health	Integer	1-5	Health consequence score
environment	Integer	1-5	Environment consequence score
production	Integer	1-5	Production consequence score
operating_cost	Integer	1-5	Operating cost score
capital_cost	Integer	1-5	Capital cost score
schedule	Integer	1-5	Schedule impact score
revenue	Integer	1-5	Revenue impact score
communications	Integer	1-5	Communications impact score
compliance	Integer	1-5	Compliance impact score
reputation	Integer	1-5	Reputation impact score
probability	Integer	1-5	Probability of occurrence

F.3 Template 03: 03_failure_modes.xlsx

FMEA failure modes and causes (15 fields + reference sheet).

Field	Type	Validation	Description
equipment_tag	String	required	Equipment tag reference
function_description	String	required	Function description
function_type	String	dropdown: FunctionType	PRIMARY, SECONDARY, PROTECTIVE
failure_type	String	dropdown: FailureType	TOTAL or PARTIAL
what_component	String	required	Component that fails (uppercase, singular)
mechanism	String	dropdown: Mechanism	Failure mechanism (18 values)
cause	String	dropdown: Cause	Failure cause (must be valid 72-combo pair)
failure_pattern	String	dropdown: FailurePattern	Nowlan-Heap pattern (A-F)
failure_consequence	String	dropdown: FailureConsequence	Consequence classification
evidence	String	required	Observable evidence of failure
downtime_hours	Float	>=0	Estimated downtime
detection_method	String	-	How failure is detected
rpn_severity	Integer	1-10	RPN severity score
rpn_occurrence	Integer	1-10	RPN occurrence score
rpn_detection	Integer	1-10	RPN detection score

Additional sheet: "Valid FM Combinations" -- all 72 valid Mechanism+Cause pairs.

F.4 Template 04: 04_maintenance_tasks.xlsx

Maintenance task definitions (multi-sheet: Tasks, Task_Labour, Task_Materials, Task_Tools).

Sheet 1: Tasks (14 fields)

Field	Type	Validation	Description
task_id	String	required, PK	Unique task identifier
task_name	String	required, max 72 chars	Task name (SAP compatible)
task_name_fr	String	-	Task name (French)
task_type	String	dropdown: TaskType	INSPECT, CHECK, TEST, LUBRICATE, CLEAN, REPLACE, REPAIR, CALIBRATE
constraint	String	dropdown: TaskConstraint	ONLINE, OFFLINE, TEST_MODE
access_time_hours	Float	>=0	Access/setup time
budgeted_as	String	dropdown: BudgetType	NOT_BUDGETED, REPAIR, REPLACE
budgeted_life	Float	-	Budgeted life value
budgeted_life_time_units	String	-	Time-based units
budgeted_life_operational_units	String	-	Operational units
consequences	String	-	Failure consequences
justification	String	-	Task justification
origin	String	-	Task origin reference
notes	String	-	Additional notes

Note: Tasks do NOT have frequency or acceptable limits. Those belong to the maintenance strategy (Template 14).

Sheet 2: Task_Labour (8 fields)

Field	Type	Validation	Description
labour_id	String	required, PK	Unique labour line ID
task_id	String	required, FK → Tasks	Parent task reference
worker_id	String	-	Worker identifier
specialty	String	dropdown: LabourSpecialty	Worker specialty
quantity	Integer	>=1	Number of workers
hours_per_person	Float	>=0	Hours per person
hourly_rate_usd	Float	>=0	Hourly rate (USD)
company	String	-	Contractor company

Sheet 3: Task_Materials (10 fields)

Field	Type	Validation	Description
material_line_id	String	required, PK	Unique material line ID
task_id	String	required, FK → Tasks	Parent task reference
material_code	String	required	SAP material code
description	String	required	Material description
manufacturer	String	-	Material manufacturer
part_number	String	-	Part number
quantity	Float	>=0	Quantity required
unit_of_measure	String	dropdown: UnitOfMeasure	Unit
unit_price_usd	Float	>=0	Unit price (USD)
equipment_bom_ref	String	-	BOM reference

Sheet 4: Task_Tools (5 fields)

Field	Type	Validation	Description
tool_line_id	String	required, PK	Unique tool line ID
task_id	String	required, FK → Tasks	Parent task reference
item_type	String	-	TOOL or SPECIAL_EQUIPMENT
tool_code	String	required	Tool code
description	String	-	Tool description

F.5 Template 05: 05_work_packages.xlsx

Work package groupings (11 fields).

Field	Type	Validation	Description
wp_name	String	required, max 40, ALL CAPS	Work package name
wp_code	String	required	WP code
equipment_tag	String	required	Equipment tag reference
frequency_value	Float	>=0	Interval value
frequency_unit	String	dropdown: FrequencyUnit	Interval unit
constraint	String	dropdown: WPConstraint	ONLINE or OFFLINE
wp_type	String	dropdown: WPType	STANDALONE, SUPPRESSIVE, SEQUENTIAL
access_time_hours	Float	>=0	Access/setup time
task_ids_csv	String	required	Comma-separated task IDs
estimated_total_hours	Float	>=0	Total estimated hours
crew_size	Integer	>=1	Crew size

F.6 Template 06: 06_work_order_history.xlsx

Historical work order data (16 fields).

Field	Type	Validation	Description
wo_id	String	required	Work order ID
order_type	String	dropdown: WorkOrderType	Order type
equipment_tag	String	required	Equipment tag
sap_func_loc	String	-	SAP functional location
priority	String	dropdown: Priority	Priority level
description	String	required	WO description
created_date	Date	YYYY-MM-DD	Creation date
planned_start	Date	YYYY-MM-DD	Planned start
planned_end	Date	YYYY-MM-DD	Planned end
actual_start	Date	YYYY-MM-DD	Actual start
actual_end	Date	YYYY-MM-DD	Actual end
duration_hours	Float	>=0	Duration (hours)
status	String	dropdown: WOHistoryStatus	WO status
cost_labour_usd	Float	>=0	Labour cost
cost_materials_usd	Float	>=0	Materials cost
failure_found	String	-	Failure found (Y/N or description)

F.7 Template 07: 07_spare_parts_inventory.xlsx

Spare parts catalog (16 fields).

Field	Type	Validation	Description
material_code	String	required	SAP material code
description	String	required	Part description
manufacturer	String	-	Manufacturer
part_number	String	-	Part number
ved_class	String	dropdown: V,E,D	VED criticality
fsn_class	String	dropdown: F,S,N	FSN consumption
abc_class	String	dropdown: A,B,C	ABC cost class
quantity_on_hand	Integer	>=0	Current stock
min_stock	Integer	>=0	Minimum stock level
max_stock	Integer	>=0	Maximum stock level
reorder_point	Integer	>=0	Reorder point
lead_time_days	Integer	>=0	Lead time (days)
unit_cost_usd	Float	>=0	Unit cost (USD)
unit_of_measure	String	dropdown: UnitOfMeasure	Unit
applicable_equipment_csv	String	-	Comma-separated equipment tags
warehouse_location	String	-	Storage location

F.8 Template 08: 08_shutdown_calendar.xlsx

Planned shutdown schedule (9 fields).

Field	Type	Validation	Description
plant_id	String	required	Plant identifier
shutdown_name	String	required	Shutdown name
shutdown_type	String	dropdown: ShutdownType	MAJOR, MINOR, TURNAROUND
planned_start	Date	YYYY-MM-DD	Planned start
planned_end	Date	YYYY-MM-DD	Planned end
planned_hours	Float	>=0	Planned duration (hours)
areas_csv	String	-	Affected areas (comma-separated)
description	String	-	Shutdown description
work_orders_csv	String	-	Linked WO IDs (comma-separated)

F.9 Template 09: 09_workforce.xlsx

Workforce and specialty data (9 fields).

Field	Type	Validation	Description
worker_id	String	required	Worker identifier
name	String	required	Worker name
specialty	String	dropdown: LabourSpecialty	Worker specialty
shift	String	dropdown: ShiftType	Shift assignment
plant_id	String	required	Plant identifier
available	Boolean	TRUE/FALSE	Availability flag
certifications_csv	String	-	Certifications (comma-separated)
phone	String	-	Phone number
email	String	-	Email address

F.10 Template 10: 10_field_capture.xlsx

Field observation records (7 fields).

Field	Type	Validation	Description
technician_id	String	required	Technician ID
capture_type	String	dropdown: CaptureType	Capture type
language	String	dropdown: Language	Input language
equipment_tag	String	required	Equipment tag
location_hint	String	-	Location hint
raw_text	String	required	Raw observation text
timestamp	DateTime	ISO 8601	Capture timestamp

F.11 Template 11: 11_rca_events.xlsx

Root cause analysis events (10 fields).

Field	Type	Validation	Description
event_description	String	required	Event description
plant_id	String	required	Plant identifier
equipment_tag	String	required	Equipment tag
level	String	dropdown: RCALevel	GREEN, YELLOW, RED
max_consequence	String	-	Maximum consequence type
frequency	String	-	Event frequency
event_date	Date	YYYY-MM-DD	Event date
downtime_hours	Float	>=0	Downtime caused
production_loss_tonnes	Float	>=0	Production loss (tonnes)
direct_cost_usd	Float	>=0	Direct cost (USD)

F.12 Template 12: 12_planning_kpi_input.xlsx

Planning KPI input data (22 fields).

Field	Type	Validation	Description
plant_id	String	required	Plant identifier
period_start	Date	YYYY-MM-DD	Period start date
period_end	Date	YYYY-MM-DD	Period end date
wo_planned	Integer	>=0	Work orders planned
wo_completed	Integer	>=0	Work orders completed
manhours_planned	Float	>=0	Man-hours planned
manhours_actual	Float	>=0	Man-hours actual
pm_planned	Integer	>=0	PM orders planned
pm_executed	Integer	>=0	PM orders executed
backlog_hours	Float	>=0	Backlog hours
weekly_capacity_hours	Float	>=0	Weekly capacity
corrective_count	Integer	>=0	Corrective WO count
total_wo	Integer	>=0	Total WO count
schedule_compliance_planned	Integer	>=0	Planned schedule items
schedule_compliance_executed	Integer	>=0	Executed schedule items
release_horizon_days	Integer	>=0	Release horizon (days)
pending_notices	Integer	>=0	Pending notices
total_notices	Integer	>=0	Total notices
scheduled_capacity_hours	Float	>=0	Scheduled capacity
total_capacity_hours	Float	>=0	Total capacity
proactive_wo	Integer	>=0	Proactive WO count
planned_wo	Integer	>=0	Planned WO count

F.13 Template 13: 13_de_kpi_input.xlsx

Defect elimination KPI data (13 fields).

Field	Type	Validation	Description
plant_id	String	required	Plant identifier
period_start	Date	YYYY-MM-DD	Period start date
period_end	Date	YYYY-MM-DD	Period end date
events_reported	Integer	>=0	Events reported
events_required	Integer	>=0	Events requiring analysis
meetings_held	Integer	>=0	DE meetings held
meetings_required	Integer	>=0	DE meetings required
actions_implemented	Integer	>=0	Actions implemented
actions_planned	Integer	>=0	Actions planned
savings_achieved	Float	>=0	Savings achieved (USD)
savings_target	Float	>=0	Savings target (USD)
failures_current	Integer	>=0	Current period failures
failures_previous	Integer	>=0	Previous period failures

F.14 Template 14: 14_maintenance_strategy.xlsx

Maintenance strategy linking failure modes to tasks with intervals and limits (31 fields + 2 reference sheets).

*This is the **core strategy template** that connects the FMEA analysis (Template 03) to maintenance tasks (Template 04). Each row represents a strategy record: one failure mode → one or two maintenance tasks, with frequency, acceptable limits, and justification.*

Field	Type	Validation	Description
strategy_id	String	required	Unique strategy identifier
equipment_tag	String	required	Equipment tag reference
maintainable_item	String	required	Maintainable item name
function_and_failure	String	required	Function and failure description
what	String	required	FM "what" component
mechanism	String	dropdown: Mechanism	Failure mechanism (18 values)
cause	String	dropdown: Cause	Failure cause (72-combo validated)
status	String	dropdown: ACTIVE, DRAFT, ELIMINATED	Strategy status
tactics_type	String	dropdown: StrategyType	CBM, TBM, FBM, RTF, FFI, REDESIGN
primary_task_id	String	required, FK → Tasks	Primary task reference
primary_task_interval	Float	>=0	Primary task interval value
operational_units	String	-	Operational interval units
time_units	String	dropdown: FrequencyUnit	Time-based interval units
primary_task_acceptable_limits	String	-	Measurable acceptance criteria
primary_task_conditional_comments	String	-	Conditional notes
primary_task_constraint	String	dropdown: TaskConstraint	ONLINE, OFFLINE, TEST_MODE
primary_task_task_type	String	dropdown: TaskType	Primary task type
primary_task_access_time	Float	>=0	Primary task access time
secondary_task_id	String	FK → Tasks	Secondary task reference
secondary_task_constraint	String	dropdown: TaskConstraint	Secondary task constraint
secondary_task_task_type	String	dropdown: TaskType	Secondary task type
secondary_task_access_time	Float	>=0	Secondary task access time
secondary_task_comments	String	-	Secondary task notes
budgeted_as	String	dropdown: BudgetType	NOT_BUDGETED, REPAIR, REPLACE
budgeted_life	Float	-	Budgeted life value
budgeted_life_time_units	String	-	Budgeted life time units
budgeted_life_operational_units	String	-	Budgeted life operational units
existing_task	String	-	Reference to existing task (if any)
justification_category	String	dropdown: JustificationCategory	Justification category
justification	String	-	Strategy justification text
notes	String	-	Additional notes

Additional sheet 1: "Valid FM Combinations" -- all 72 valid Mechanism+Cause pairs.

Additional sheet 2: "Strategy Type Rules" -- 6 maintenance strategy types with selection criteria:

Strategy	When to Use
CBM	Condition-Based: detectable P-F interval, feasible monitoring
TBM	Time-Based: age-related pattern, known useful life
FBM	Failure-Based: redesign not justified, low consequence RTF
RTF	Run-to-Failure: non-critical, no safety/environmental consequence
FFI	Failure-Finding Interval: hidden failure, protective function
REDESIGN	Redesign: no applicable task, unacceptable risk

End of variable listing.